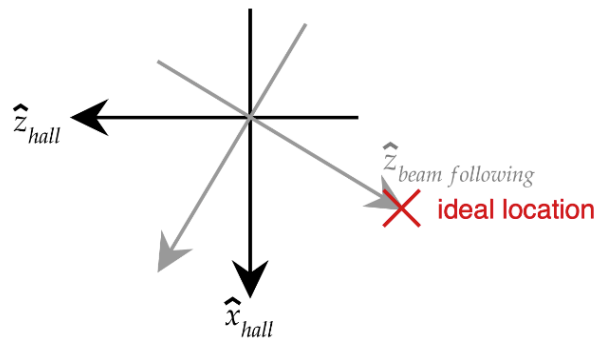


LAD_hodowall position

	CEBAF Coord. System			Beam Following Coord. System			dYaw	dPitch	dRoll
	X[m]	Y[m]	Z[m]	dx[mm]	dy[mm]	dz[mm]			
LAD1	-123.74601	99.99568	-381.76705	14.63	-1.82	32.30	-0.1800	-0.2848	-0.0547
LAD3	-128.44342	100.00260	-383.48154	3.39	5.10	-1.60	0.1046	0.0877	-0.0077
LAD4	-126.12369	100.00294	-382.81534	15.38	5.44	41.97	-0.2597	-0.4985	-0.3552
LAD5	-123.77851	99.99752	-382.16368	34.29	0.02	41.23	0.1938	-0.4736	-0.1558
LAD6	-126.00501	100.01053	-383.21780	22.32	13.03	28.83	0.0012	-0.1922	-0.3776

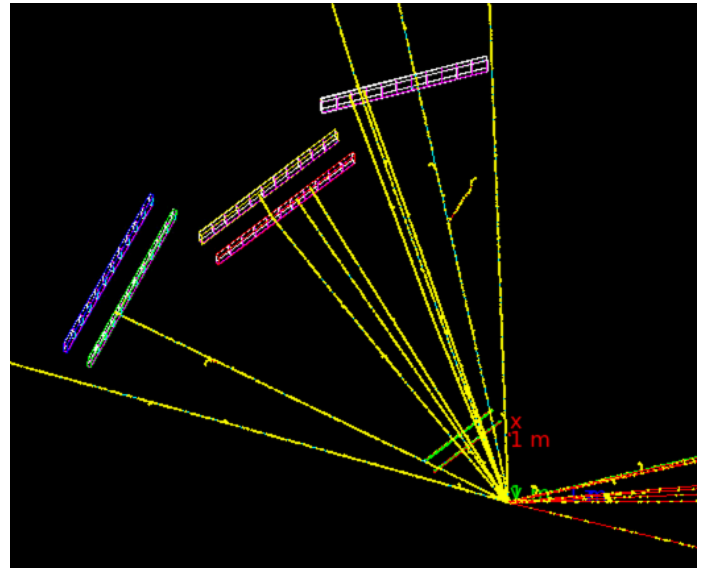
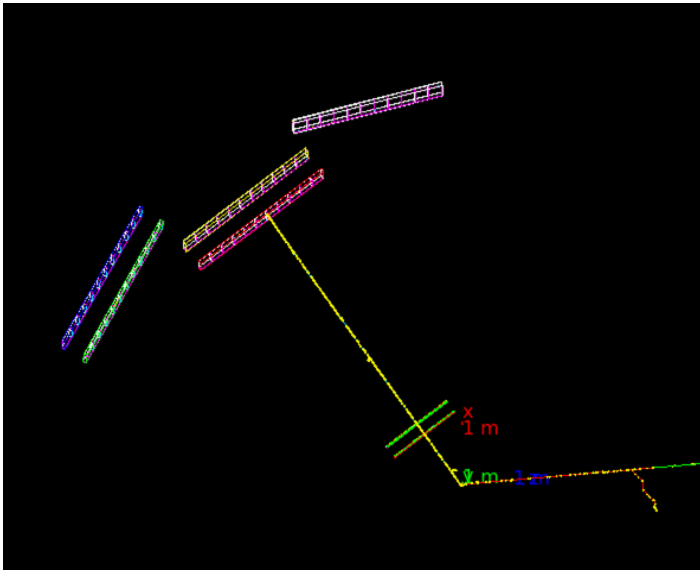
- Beam Following(dx, dy, dz, dPitch, dYaw, dRoll) -> Hall C:



$$\begin{pmatrix} dx_{hall} \\ dy_{hall} \\ dz_{hall} \\ dPitch_{hall} \\ dYaw_{hall} \\ dRoll_{hall} \end{pmatrix} = \begin{pmatrix} \cos \theta & 0 & \sin \theta & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ -\sin \theta & 0 & \cos \theta & 0 & 0 & 0 \\ 0 & 0 & 0 & \cos \theta & 0 & \sin \theta \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -\sin \theta & 0 & \cos \theta \end{pmatrix} \begin{pmatrix} dx \\ dy \\ dz \\ dPitch \\ dYaw \\ dRoll \end{pmatrix}$$

- Each wall has different theta!!!

Plane	theta/ deg	dx_hall/m	dy_hall/m	dz_hall/m	dPitch_hall /deg	dYaw_hall /deg	dRoll_hall /deg
LAD1	150	0.00348	-0.00182	-0.035288	0.219294	-0.18	0.189772
LAD3	104	-0.002373	0.0051	-0.002902	-0.028688	0.1046	-0.083232
LAD4	127	0.024263	0.00544	-0.037541	0.016329	-0.2597	0.611884
LAD5	150	-0.009081	0.00002	-0.052851	0.33225	0.1938	0.371727
LAD6	127	0.009592	0.01303	-0.035176	-0.185896	0.0012	0.380743



- Calculate the (x,y,z) in Hall Coord. System
 - Beam Following → Hall corrections:

Plane	theta (deg)	Distance (m)	x_hall (m)	y_hall (m)	z_hall (m)	dPitch (deg)	dYaw (deg)	dRoll (deg)
LAD1	150.12723	6.58832	3.28148	-0.00182	-5.71295	0.219294	-0.18	0.189772
LAD3	104.03159	6.1484	5.96495	0.0051	-1.49072	-0.028688	0.1046	-0.083232
LAD4	127.1552	5.67799	4.52537	0.00544	-3.42937	0.016329	-0.2597	0.611884
LAD5	150.31733	6.19132	3.06592	0.00002	-5.37891	0.33225	0.1938	0.371727
LAD6	127.24318	5.25889	4.18646	0.01303	-3.18267	-0.185896	0.0012	0.380743

Definitions

- x_{hall} , y_{hall} , and z_{hall} are the actual wall-center coordinates in the Hall coordinate system:
- $x_{\text{hall}} = x_{\text{theory}} + dx_{\text{hall}}$, with equivalent expressions for y and z .
- **Distance** is the three-dimensional distance from the Hall origin to the wall center:

$$d = \sqrt{x_{\text{hall}}^2 + y_{\text{hall}}^2 + z_{\text{hall}}^2}$$
- **theta** is the angular position of the wall center relative to the Hall $+Z$ axis:

$$\theta = \text{atan2}(x_{\text{hall}}, z_{\text{hall}})$$
- These three angular corrections change only the wall orientation. The wall is rotated about its own center, so its center coordinates remain unchanged.
 - **dPitch** is the rotation of an axis through the wall center and parallel to the Hall X axis.
 - **dYaw** is the rotation of an axis through the wall center and parallel to the Hall Y axis.
 - **dRoll** is the rotation of an axis through the wall center and parallel to the Hall Z axis.

